



CSCI 232

232 Algorithms & Data Structures

LECTURE: 1

J. L. Pach

OUTLINE:

- ✂ Syllabus, Textbook, Canvas
- ✂ Introduction
- ✂ History of C
- ✂ Positional notation
- ✂ IDE
- ✂ Hello World
- ✂ Compiler vs Interpreter

SYLLABUS

SOME BASIC FACTS ABOUT THE COURSE

Course Name:

⌘ Programming with C (CSCI 112)

Credit Hours:

⌘ 3 credits

⌘ 1 hour lecture twice a week

⌘ 3 hours lab per week

Lecture (Mondays and Fridays)

⌘ 12:00 – 12:50 PM.

⌘ Science & Engineering Building (S&E)
308

Lab (Mondays)

⌘ 2:00 – 4:50 PM

⌘ Engineering Lab/Classroom Building
(ELC) 315

SYLLABUS

- ✂ Course Description
- ✂ Textbooks
- ✂ Class Rules
- ✂ Grading
- ✂ Accommodations & Academic Dishonesty
- ✂ Declaration of authorship

COURSE DESCRIPTION

This course provides a comprehensive introduction to structured programming using the C language. Student will gain a deep understanding of memory management techniques such as pointers and dynamic allocation. The skills acquired in this course will be essential for those who wish to pursue further studies in languages like C++, C#, and Java, as well as microcontroller programming. Additionally, this course will lay a strong foundation for understanding computer architecture.

...A FEW WORDS

This course is designed for students in:

- ✂ Computer Science (CS),
- ✂ Software Engineering (SE),
- ✂ Electronics programs.

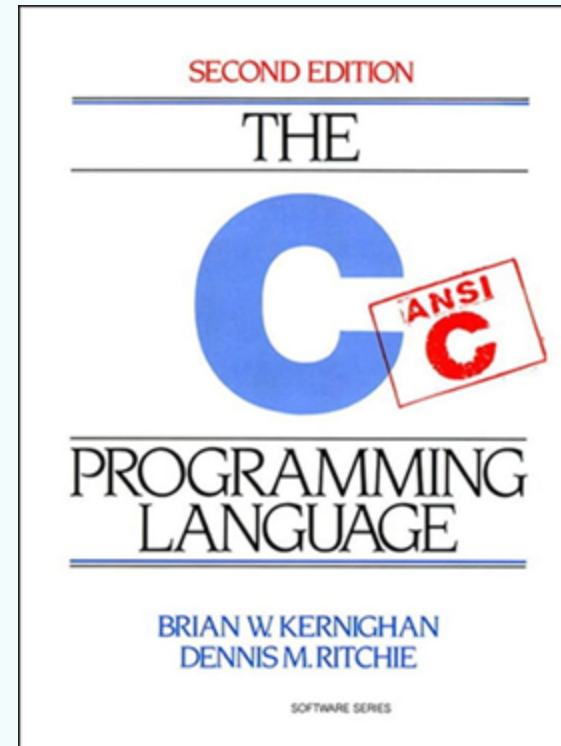
No prior programming experience is required — only basic computer literacy, such as using a web browser, email, and navigating the MS Windows environment.

TEXTBOOKS

Brian W. Kernighan, Dennis M. Ritchie. C Programming Language, 2nd Edition. Prentice Hall, 1988



Dennis M. Ritchie



ANSI(C89) VS C99

Last year, the course was taught using the oldest and most widely adopted ANSI standard — C89. This choice was made because C89 does not include many of the  conveniences introduced in later versions. While these conveniences make life easier for professional programmers, they can make it harder for beginners to fully grasp the fundamentals of programming in C.

The standard now used in our classes is **C99**, which is widely applied in embedded systems, especially in the field of electronics. This standard significantly lowers the  entry barrier for beginners (for example, allowing variable declarations inside a for loop and supporting single-line comments `//`), while avoiding the complex and sometimes confusing features introduced in newer standards.

COURSE RULES

1. This course is an in-person course.
2. Attendance at every class is required.
3. Excused Absences. If there is any medical or any other kind of emergency, please let the instructor know immediately.
4. Makeup exams will only be given if you bring a valid medical documentation.
5. Each task without a sample required comment (below) is worth zero points.

GRADING

The course grade will be determined by two equally weighted components:

- ✂ the lecture component (40%)
- ✂ the laboratory component (60%).

The laboratory component will be evaluated based on three criteria:

- ✂ entrance quiz (30%),
- ✂ assignment (40%),
- ✂ brief concluding quiz (30%).

LABORATORY RULES

Entrance Quiz (optional):

If included, the entrance quiz will consist of 3 questions. To pass, a student must score  at least 2 points. Failed entrance quizzes may be retaken up to two times during the semester. Only failed quizzes can be retaken.

Assignment:

 Students will have 6 days to complete and submit their assignment.

Brief Concluding Quiz (optional):

 If included, this quiz serves as a summative assessment component, reinforcing the material covered during the lab session.

DETAILS

Accommodations:

- ✚ Students who need any type of accommodation should work with Montana Tech Disability Services and provide appropriate documentation as soon as possible.

Academic Dishonesty:

- ✚ You are encouraged to work in teams and use many resources including books and the Internet. However, each student must turn in his/her own work, and each student is responsible for understanding anything that is turned in. Refer to the Student Conduct Code for more information regarding plagiarism and cheating.

DECLARATION OF AUTHORSHIP

I acknowledge that I have worked on this assignment independently, except where explicitly noted and referenced. Any collaboration or use of external resources has been properly cited. I am fully aware of the consequences of academic dishonesty and agree to abide by the university's academic integrity policy. I understand the importance the consequences of plagiarism.

COMMENTS

There must be 4 lines of comments at the top of each source file. This heading should include your name, the class and semester, and the assignment number, and required statement. Source file without this comment gets **zero** points.

Template

```
//Jakub Pach  
//CSCI 112 Fall 2026  
//Programming Assignment #1  
//I acknowledge that I have worked on this assignment independently, except where explicitly noted and referenced...
```



Lecture & Laboratory:

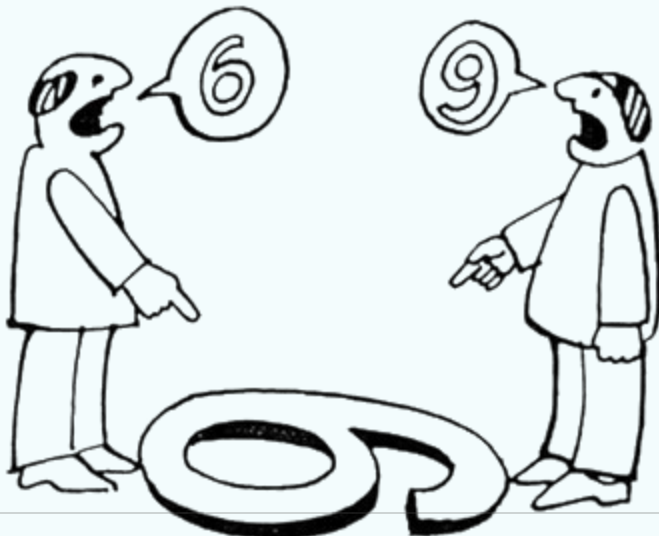
(76078) CSCI 112 SEC 01 :PROGRAMMING WITH C

Check if you are registered for the course and have access to it!

INTRODUCTION

MEANING OF WORDS

Sometimes, when we look at the same thing, we see entirely different things. Therefore, to avoid misunderstandings, I will spend a considerable amount of time clarifying the nuances associated with translating the meanings of the concepts we will be using in our classes. All the material presented in this course will be of practical use.



COMPILER

1. A compiler takes the entire source code and translates it into a machine code file, often called **an executable**.
2. This executable file contains instructions that the computer's processor can directly execute.
3. Once compiled, the program can run independently without the need for the original source code or a compiler.

INTERPRETER

1. An interpreter translates the source code **line by line** as the program is running.
2. It doesn't create a separate executable file. Instead, it uses a virtual machine to execute the translated code.
3. The virtual machine provides an environment that mimics a real computer, allowing the program to run even if the underlying hardware architecture is different.

COMPILER VS INTERPRETER

Think of a compiler as a translator who translates an entire book from one language to another before you start reading it. An interpreter, on the other hand, is a translator who translates each sentence as you read it. A compiler translates the entire program at once, while an interpreter translates it line by line.

HISTORY OF C

Similarly, in music, we first learn to read sheet music, and only after mastering the basics can we investigate into the history and evolution of musical notation. In this course, we'll follow a similar approach. We'll start by learning how to write code in C, and we'll save the history of the C language for later, once you have gained significant experience. Then, we can explore the origins of certain language constructs.

BIT & BYTE

BIT & BYTE

A **bit** is the smallest unit of data in a computer, representing a single binary value: either a **0** or a **1**.

A **byte** is a group of eight bits. A single byte can represent a wide range of values, such as a single character (like the letter '**A**' or the symbol '@') or an integer from **0 to 255**.

POSITIONAL NOTATION

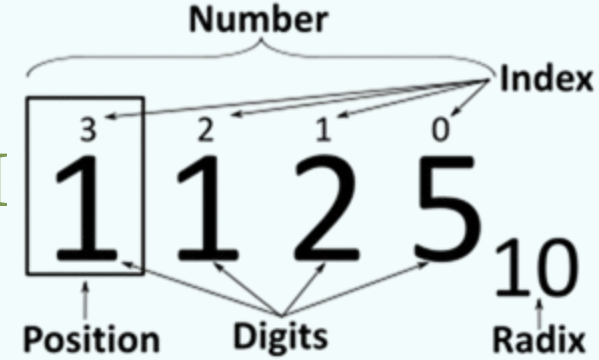
POSITIONAL NOTATION

✂ Usually means the extension to any base (2, ..., 8,10,16 etc.) of the Hindu–Arabic numeral system (or decimal system).

✂ A positional system is a numeral system in which the contribution of a digit to the value of a number is the value of the digit multiplied by a factor determined by the position of the digit.

✂ In modern positional systems, such as the decimal system, the position of the digit means that its value must be multiplied by some value: in 444, the three identical symbols represent four hundreds, four tens, and four units, respectively, due to their different positions in the digit string.

BASE OF THE NUMERAL SYSTEM

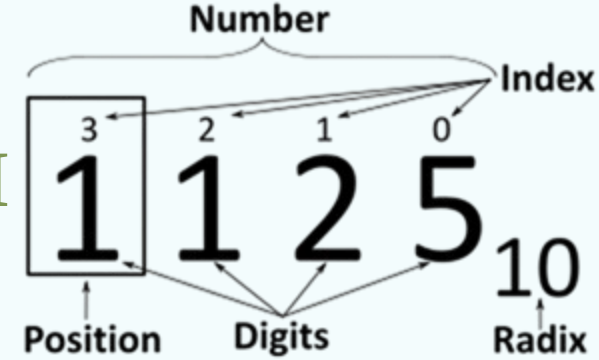


✂ In mathematical numeral systems the radix r is usually the number of unique digits, including zero, that a positional numeral system uses to represent numbers.

The highest symbol of a positional numeral system usually has the value one less than ✂ the value of the radix of that numeral system. The standard positional numeral systems differ from one another only in the base they use.

✂ The radix is an integer that is greater than 1, since a radix of zero would not have any digits, and a radix of 1 would only have the zero digit.

BINARY NUMERAL SYSTEM



$$A = \sum_{i=0}^{n-1} R^i \cdot d_i$$

$$\overset{2}{1} \overset{1}{2} \overset{0}{3}_{(10)} = (10^2 \cdot 1) + (10^1 \cdot 2) + (10^0 \cdot 3)$$

$$\overset{7}{0} \overset{6}{1} \overset{5}{1} \overset{4}{1} \overset{3}{1} \overset{2}{0} \overset{1}{1} \overset{0}{1}_{(2)} = (2^7 \cdot 0) + (2^6 \cdot 1) + (2^5 \cdot 1) + (2^4 \cdot 1) + (2^3 \cdot 1) + (2^2 \cdot 0) + (2^1 \cdot 1) + (2^0 \cdot 1) = 123_{(10)}$$

CONVERTING DECIMAL NUMBERS TO BINARY

Steps:

1. Divide the decimal number by 2. The remainder of the division will be either 0 or 1.
2. If the remainder is 0, write down a 0.
3. If the remainder is 1, write down a 1.
4. Repeat steps 2 and 3 until the decimal number is 0.
5. Read the numbers **from bottom to top** to get the binary number.

EXAMPLE

123_{10}

✂ $123 / 2 = 61$ (remainder 1)

✂ Write down a 1.

✂ $61 / 2 = 30$ (remainder 1)

✂ Write down a 1.

✂ $30 / 2 = 15$ (remainder 0)

✂ Write down a 0.

✂ $15 / 2 = 7$ (remainder 1)

✂ Write down a 1.

✂ $7 / 2 = 3$ (remainder 1)

✂ Write down a 1.

✂ $3 / 2 = 1$ (remainder 1)

✂ Write down a 1.

✂ $1 / 2 = 0$ (remainder 1)

✂ Write down a 1.

CONVERTING BINARY NUMBERS TO OCT

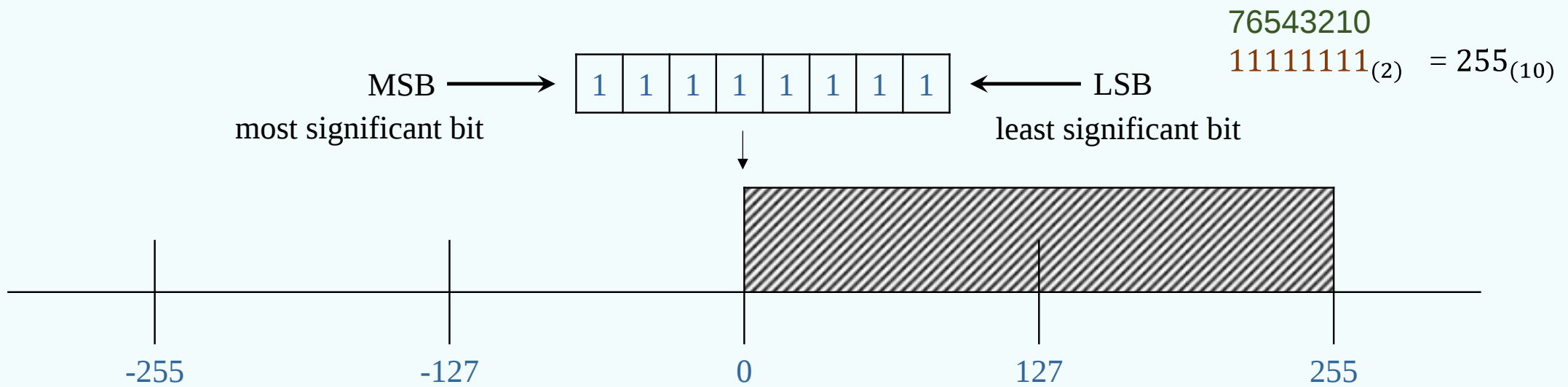
1111011_2

$001.111.011 = 1.7.3$

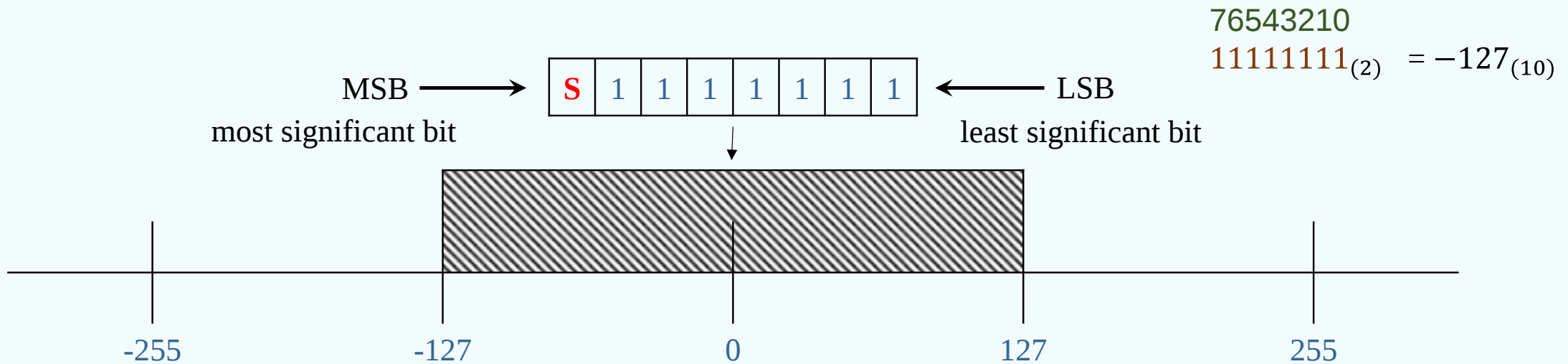
$0111.1011 = 7.B$

Radix	Value	Binary Value	Symbol
Octal	0	0000	0
Octal	1	0001	1
Octal	2	0010	2
Octal	3	0011	3
Octal	4	0100	4
Octal	5	0101	5
Octal	6	0110	6
Octal	7	0111	7
Hexadecimal	8	1000	8
Hexadecimal	9	1001	9
Hexadecimal	10	1010	A
Hexadecimal	11	1011	B
Hexadecimal	12	1100	C
Hexadecimal	13	1101	D
Hexadecimal	14	1110	E
Hexadecimal	15	1111	F

UNSIGNED REPRESENTATION



SIGN-MAGNITUDE REPRESENTATION



DATA IN A COMPUTER

can essentially be stored using two standards:

1. integers represented in binary,
2. real (floating-point) numbers stored according to the IEEE 754 standard.

Everything else is a combination or interpretation based on these two fundamental forms of representation.

IDE

Integrated development environment

INTEGRATED DEVELOPMENT ENVIRONMENT

An Integrated Development Environment (IDE) is a comprehensive software application that consolidates the essential tools needed for software development into a single graphical user interface. It typically combines:

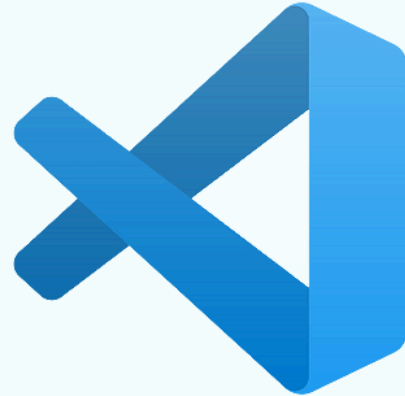
1. a source **code editor** for writing code
2. a **compiler** and **linker** to transform that code into an executable program.
3. a built-in **debugger** to help developers identify and resolve logical errors

efficiently during the programming process.

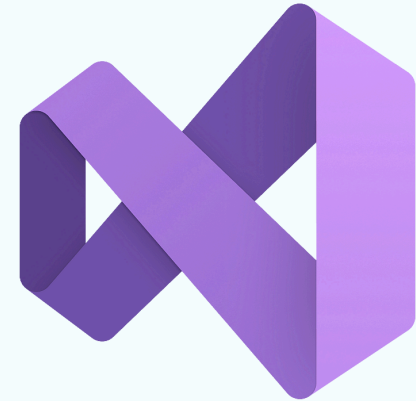
INTEGRATED DEVELOPMENT ENVIRONMENT



- Code::Blocks



- Visual Studio Code



- Visual Studio

THANK

You